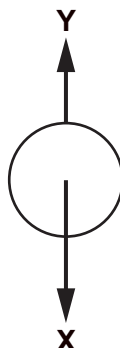


5. A ball is dropped from rest.
Two forces, X and Y, act on the ball during its fall.



- (a) (i) Name the forces X and Y. [2]

X Y

- (ii) State what happens to the size of the forces, if anything, as the ball speeds up. [2]

Force X

Force Y

- (b) Complete the following sentences by underlining the correct phrase in the brackets. [2]

- (i) The ball speeds up at the start of its fall because force Y is (**smaller than** / **equal to** / **larger than**) force X.
- (ii) The ball eventually reaches a constant speed because force Y is (**smaller than** / **equal to** / **larger than**) force X.

- (c) After being dropped from rest, the ball takes 2 seconds to hit the ground.

- (i) Use the equation:

$$v = u + at$$

to calculate the final speed, v , with which the ball hits the ground.
(Acceleration due to gravity, $g = 10 \text{ m/s}^2$)

[2]

$v =$ m/s



(ii) Use an equation from page 2 and your answer to part (c)(i) to calculate the height, x , from which the ball was dropped.

[3]

$x = \dots\dots\dots$ m

Examiner
only

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